

Multimedia Databases Exercises SS 2026

University of Passau

Prof. Dr. Mario Döller, Prof. Dr. Harald Kosch, Manuel Lehner

Exercise 4

Subject: Image Processing Part 1

Release Date: May 7, 2026

Task 1: Image Formats

- a) Describe the differences between vectorial and raster (bitmap) image formats.
- b) Outline the main properties of the following image formats: GIF, PNG and JPEG with respect to the following dimensions: color model, color depth, compression, file size.

Task 2: Image Processing and Resolution

Let I be an 12 x 12 RGB image.

- a) Consider an RGB image implementation in **Java** that stores pixels in a 3D array (of integers), where each dimension corresponds to a color channel. How much memory is required to store I?
- b) One way to reduce memory consumption is to transform the 3D array representation into a 2D array representation. A pixel is packed into a single integer (32 bit), where each channel is coded in 8 bits. The desired result is depicted in Figure 1. How much memory is saved compared to the 3D array representation?
- c) Describe how we can retrieve the individual R, G, B samples from the packed pixel.
- d) For an iPhone 17, given a screen size of 6.3 inches, an aspect ratio of 19,5 : 5, and a resolution of 2622 x 1206 pixels, calculate it's pixel density (Show the steps).

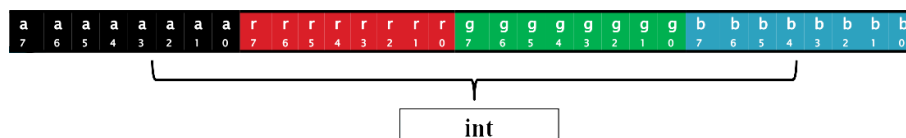


Figure 1: A pixel as single integer

Task 3: Quantization and Dithering

- a) Write down the uniform and median cut color quantization algorithms (In simple steps, not code/pseudocode).
- b) Give a brief explanation of noise dithering.